



Impact of the environmental protection law on corporate biodiversity risk attention - An empirical analysis based on data of Chinese listed companies

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ABSTRACT

China introduced the directive-based Environmental Protection Law (EPL) in 2015 to improve and protect the environment. This paper uses the data of Chinese listed companies from 2010 to 2023 to construct a difference-in-differences model and finds that the implementation of the EPL promotes biodiversity risk attention among enterprises in biodiversity-sensitive industries. Furthermore, ESG performance and executives' environmental awareness play a mediating effect in this relationship. The heterogeneity analysis reveals that the EPL has a more significant impact on the attention paid by state-owned enterprises and enterprises participating in the carbon emission rights trading pilot program to the biodiversity risk. This paper provides relevant evidence for further incorporating biodiversity risk into the enterprise assessment standards.

Introduction

According to the whitepaper "Biodiversity Conservation in China" released by the Information Office of the State Council in October 2021, China has elevated biodiversity conservation to a national strategy. Concerning the issue of biodiversity risk, studies have demonstrated that the European financial industry does not pay sufficient attention to biodiversity risks (Azizi et al., 2025). A similar pattern is observed in the Chinese context, where companies demonstrate a comparable lack of attention to biodiversity risk. The majority of these entities prioritize emissions from their direct operations, thereby neglecting the critical issue of biodiversity risk. A significant number of ESG rating agencies in China do not incorporate biodiversity into their assessment criteria, the evaluation process exclusively encompasses the environmental, social, and governance concerns of companies (Zhu and Carrasco, 2025).

Unlike environmental pollution, the risk of biodiversity loss causes irreversible damage to environment (Pereira et al., 2012). The extinction of species or the collapse of ecosystems cannot be fully restored through technical means (Bellard et al., 2012). In contrast, the risks brought about by carbon dioxide, sulfur dioxide, and other pollutants, although having far-reaching impacts, can be gradually alleviated over a longer time scale through relevant emission reduction measures (Appio et al., 2025). For instance, carbon emission reduction can be achieved through policies such as digital trade (D. Gao et al., 2025) and VAT reform (Gao et al., 2022). Furthermore, the risks associated with biodiversity exhibit a high degree of spatial heterogeneity. The absence of biodiversity poses different threats to different spatial environments, and these threats directly affect the current ecosystem in that particular space. However, for instance, the greenhouse effect caused by carbon pollution has global homogeneity and is relatively evenly distributed geographically.

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Therefore, studying the impact of enterprise activities on environmental pollution cannot completely replace the study of the effects of enterprise activities on biodiversity.

The presence of robust national governance systems has been demonstrated to enhance corporate biodiversity risk attention through the implementation of mandatory biodiversity disclosure (Orazalin et al., 2025). The sustainability element of China's Belt and Road Initiative encompasses biodiversity, and in the final two years of the Initiative, a significant number of Chinese companies have come to recognize the importance of biodiversity (He et al., 2024). It shows that the implementation of effective environmental regulations can mitigate the impact of industrial production and business activities on biodiversity to a certain extent (Zhao et al., 2025). Low-carbon policies in China can also promote a corporate focus on biodiversity, and companies that are more biodiversity-conscious can increase their access to finance (Nie et al., 2025) (Nie and Zhang, 2025). The pilot policy of the national park system as a biodiversity conservation strategy also has the potential to significantly improve the corporate environmental information disclosure (Chen et al., 2025). The pilot policies for intelligent manufacturing can also promote the green development of enterprises (D. Gao et al., 2025).

This heightened focus can positively impact the enterprise's risk aversion and resource utilization efficiency (Wang and Xue, 1998). The disclosure of relevant information engendered by elevated biodiversity risk attention has the potential to enable companies to benefit from various aspects, such as gaining market recognition, consumer trust, and increasing corporate exposure (Zhao and Li, 2022). Biodiversity risk attention will also have a significant impact on the corporate stock returns (Zhou et al., 2025), stock price collapse (Liang et al., 2024), corporate bankruptcy (Adamolekun, 2024).

Environmental regulations can reduce biodiversity risk by enhancing information transparency (Hu and Xu, 2025) and environmentally friendly technologies innovation (Ma et al., 2024). The "Biodiversity Conservation Strategy and Action Plan (2023–2030)" of China clearly requires that in enterprise environmental information disclosure, the impact on biodiversity should be included, and mandatory disclosure of information such as the occupation of biological habitats and the development in ecologically sensitive areas should be made. Furthermore, in the 2025 Two Sessions proposals, it was also advocated that listed companies disclose the "Corporate Biodiversity Impact Index", and habitat protection was included in the key performance indicators. Similarly, "EU Deforestation Regulation, EUDR" influences biodiversity through the transmission of the supply chain (de Oliveira et al., 2024). This regulation prohibits the export of agricultural products related to deforestation, requires enterprises to trace the source of their raw materials, and actively promotes the development of eco-friendly technologies such as bio-based materials. In addition, digital technologies such as artificial intelligence can also promote the development of renewable energy (D. Gao et al., 2025), but the uncertainty of the policies will restrain this effect (D. Gao et al., 2025).

In summary, biodiversity risks differ significantly from traditional pollutant discharge risks. Beyond addressing their own emissions, enterprises must also consider the location and scope of their activities during daily operations to avoid damaging biodiversity. Moreover, directive-based policies can more effectively enhance corporate environmental awareness under China's unique national conditions, thereby increasing corporate attention to biodiversity risk. Therefore, this study uses the Environmental Protection Law as an external policy shock and selects biodiversity-sensitive industries as the treatment group. The results indicate that the Environmental Protection Law effectively enhances corporate attention to biodiversity risks, with corporate ESG performance and executives' environmental awareness serving as mediating factors in this process. Given the large number of enterprises, the command-and-control policies are more applicable than market incentives or public participation. Therefore, studying the effectiveness of China's command-and-control policies can provide valuable policy references for developing countries.

This paper makes the following contributions. First, while prior research has primarily focused on environmental regulations' effects on corporate emissions, green technology innovation, and ESG performance, few scholars have explored their impact on elevating biodiversity risk awareness. Second, it identifies industries with significant biodiversity impacts through a rational screening process. Third, while previous literature has predominantly focused on urban-level research, this study examines whether environmental laws can promote corporate attention to biodiversity risks at the industry level, thereby filling a gap in this area.

Research methodology

2.1. Variables and data source

2.1.1. Dependent variable

This paper selects the data of Chinese listed companies from 2010 to 2023 as research sample. Refer to (He et al., 2024; Giglio et al., 2023; Giglio et al., 2024), corporate biodiversity risk attention is measured using the ratio of biodiversity related keywords¹ to the total number of words in the enterprises' annual report. The reason is the annual report largely reflects the company's strategic characteristics and development direction. The activities of the enterprise play a significant role in biodiversity conservation. The attention given to biodiversity risk in the annual report demonstrates the company's commitment to sustainable development and environmental responsibility, which is consistent with the research using the US 10k report. Considering the data availability and continuity, this paper selects the data of 1855 Chinese listed companies. As seen in Table 1 and Fig. 1, the Chinese enterprises' biodiversity risk attention has increased significantly since 2015.

¹ Biodiversity-related terms: biodiversity, ecosystem(s), ecology/ecological, habitat(s), species, (rain)forest(s), deforestation, fauna, flora, marine, tropical, freshwater, wetland, wildlife, coral, aquatic, desertification, carbon sink(s), ecosphere, and biosphere.

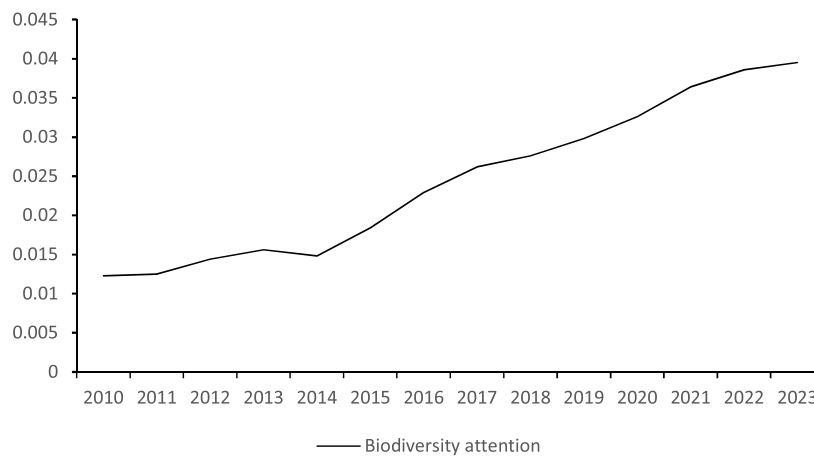


Fig. 1. The trend of average corporate biodiversity attention.

2.1.2. Independent variable

This paper constructs a difference-in-differences model to empirically analyze the impact of the EPL on corporate biodiversity risk attention, the individual fixed effect and time fixed effect are included in the model. We select the implementation of EPL in 2015 as an external shock. Select enterprises in biodiversity sensitive industries as the experimental group.² Additionally, this study employs sensitivity analysis using the ratio of percentage of carbon emissions on percentage change of revenue. The results shown in Figs. 2–4 indicate that, compared with the control group, the industry sensitivity in the treatment group is stronger. Moreover, on the whole, the mean sensitivity of the treatment group is also greater than that of the control group. Following the implementation of the Environmental Protection Law in 2015, the correlation between revenue and carbon emissions weakened, suggesting enterprises may have achieved decoupled growth.

2.1.3. Control variables

According to (He et al., 2025) and (Nie and Zhang, 2025), we select firm size, net profit margin on total assets, gearing ratio, cash and cash equivalents turnover ratio, operating income growth rate, and total green patent applications as firm-level control variables. Regional GDP, the percentage of secondary industries, and regional population as city-level control variables.

2.1.4. Mediating variables

We use HuaZheng ESG ratings data to measure ESG performance of firms. we adopt the method of (Wang et al., 2022) use the background of senior executives as a criterion to measure executives' environmental awareness.³ The text in an executive's resume can reflect their past green initiatives and, to a certain extent, gauge their level of environmental awareness.

2.2. Regression model

(Yan et al., 2025) found that ESG rating can realize the flexible disclosure of climate risks by strengthening the disclosure of environmental information of enterprises. Therefore, we believe that ESG rating can also promote enterprises to pay attention to

² In April 2022, the Center for Nature and Society at Peking University, the Shan Shui Conservation Center, Wande Technology Co., Ltd., and Huatai Securities Co., Ltd. jointly released the "Corporate Biodiversity Pressure Assessment Report (2021)". Based on the list of 55 first-level industries and 173 second-level industries proposed in the "Catalogue for the Classification Management of Environmental Impact Assessments for Construction Projects (2021 Edition)" issued by the Ministry of Ecology and Environment of China, as well as the industries of ecological impact projects listed in the "Environmental Impact Report Form for Construction Projects", they have proposed the assessment standards for enterprises' biodiversity pressure, as shown in Tables 2 and 3. And they selected some enterprises from the manufacturing industry, mining industry, accommodation and catering industry, agriculture, forestry, and fishing industry, transportation, warehousing and postal industry, construction industry, real estate industry, production and supply of electricity, heat, gas and water, water conservancy, environmental and public facilities management industry, scientific research and technical service industry for assessment. In the assessment, over 60% of the enterprises exerted pressure on the ecological environment. Among them, the high-pressure level accounted for 15%, the medium-pressure level accounted for 16%, and the low-pressure level accounted for 29%. Therefore, based on the industry codes of the China Listed Companies Association, this paper selects 9 industries as the biodiversity-sensitive industries: manufacturing, agriculture, forestry, animal husbandry and fishery, production and supply of construction, mining, scientific research and technical services, transportation, warehousing and postal services, water conservancy, environment and public facilities management.

³ An executive's environmental background refers to the inclusion of keywords related to environmental protection in their resume, such as "environment," "environmental protection," "new energy," "clean energy," "ecology," "low-carbon," "sustainable," "energy-saving," and "green." This typically indicates that the executive possesses specialized knowledge or practical experience in the environmental field.

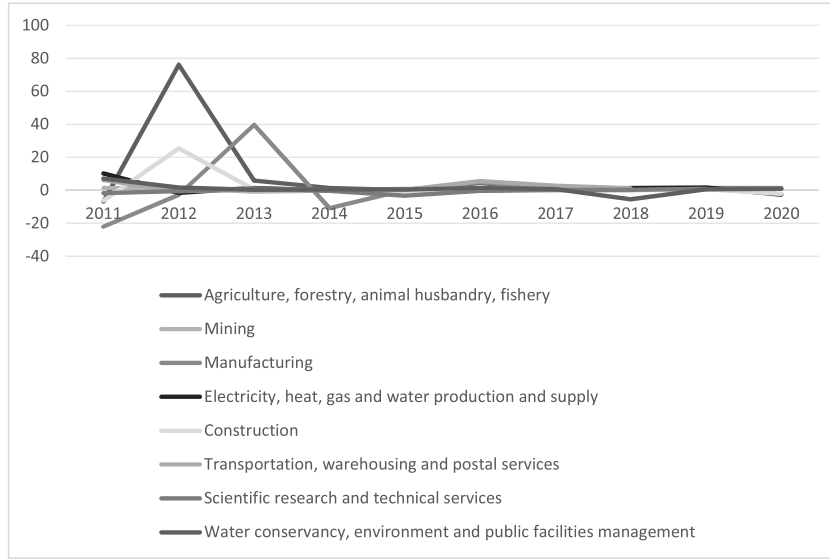


Fig. 2. Biodiversity-sensitive industries.

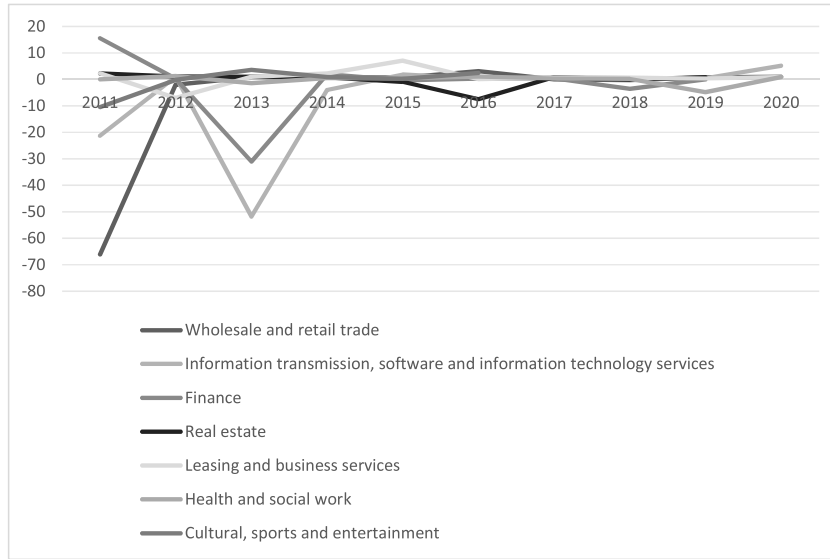


Fig. 3. Non-biodiversity-sensitive industries.

biodiversity risks and increase the disclosure of relevant information. In addition, (Zhang et al., 2025) believed that executives' green cognition can improve the performance of corporate social governance responsibility, so it will also promote the corporate biodiversity risk attention. In order to test the impact of the EPL on corporate biodiversity risk attention, and whether there is a mediating effect of corporate ESG performance and executives' environmental awareness, we construct the following empirical models:

$$BA_{it} = \alpha_0 + \alpha_1 Treat \times Post + \alpha_2 Control_{it} + \mu_i + \gamma_t + \varepsilon_{it}$$

$$M_{it} = \alpha_3 + \alpha_4 Treat \times Post + \alpha_5 Control_{it} + \mu_i + \gamma_t + \varepsilon_{it}$$

BA_{it} denotes the corporate biodiversity risk attention of firm i , year t , as an explanatory variable. α_0 denotes a constant term. $Treat \times Post$ denotes the core explanatory variable. α_1 is the coefficient for $Treat \times Post$ measuring the direction and magnitude of the impact of EPL on corporate biodiversity risk attention. $Control_{it}$ denotes the control variables. μ_i and γ_t denote individual and time fixed effects. ε_{it} denotes the random error term. M_{it} denotes the mediating variables ESG performance and executive environmental awareness.

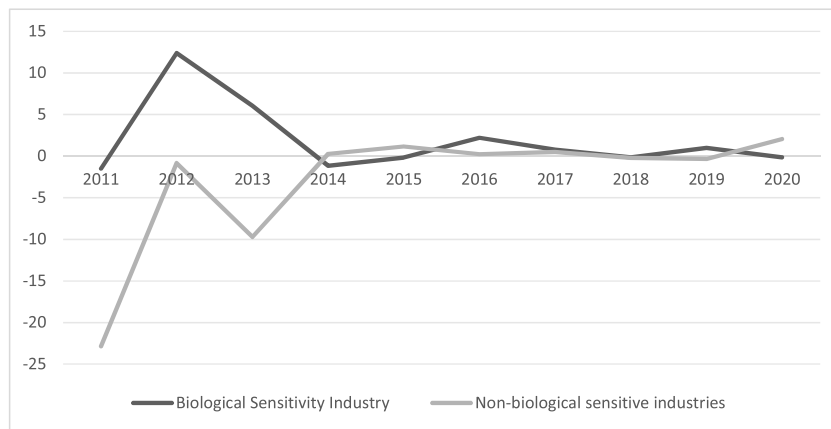


Fig. 4. Industry average comparison chart.

2.3. Data source

The data of corporate biodiversity risk attention is obtained from a published research paper (He et al., 2024), and the data of control variables is from the CSMAR database.

Empirical analysis

3.1. Descriptive statistics

Data descriptive statistics and measurement methods are shown in Table 4.

3.2. Baseline regression results

The baseline regression results are shown in Table 5. The results show that the estimated coefficient for interaction term when control variables are not included in the model is 0.00434 and it is significant at the 5 % level, indicating that the implementation of

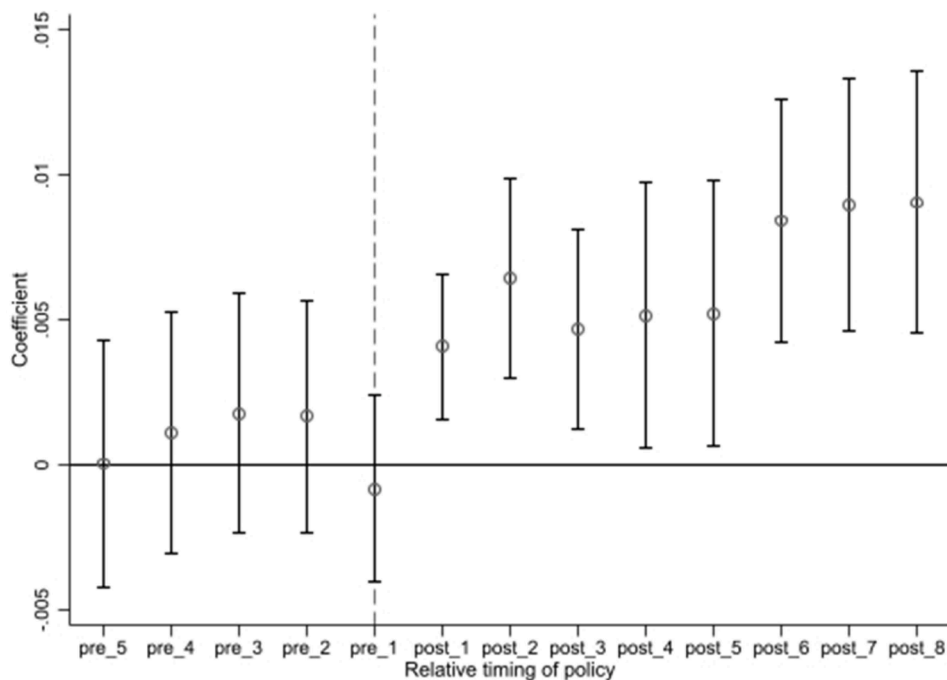


Fig. 5. Results of parallel trends test.

the EPL significantly contributes to corporate biodiversity risk attention. The estimated coefficient for interaction term after adding control variables in the model is 0.00504, and it is still significant at 5 % level, which means the empirical results of this paper are robust. The implementation of environmental laws has increased corporate biodiversity risk attention by 0.5 %. Therefore, under the EPL, enterprises will disclose more environment-related information, including biodiversity, which increases their attention about biodiversity risks.

3.3. Robustness test

3.3.1. Parallel trend test

In order to test whether the two groups converge on the trend of biodiversity risk attention before the implementation of the EPL. We construct the parallel trend test as follows:

$$BA_{it} = \alpha_0 + \sum_{k=-5}^8 \alpha_k Treat \times Post_k + \alpha_2 Control_{it} + \mu_i + \gamma_t + \varepsilon_{it}$$

$Post_k$ denotes the dummy variables for the k periods before or the k periods after the implementation of EPL. The results of the parallel trend test are shown in Fig. 5. Before the implementation of EPL, the 95 % confidence interval of estimated coefficient contains zero, indicating that there was no significant difference in biodiversity risk attention between the enterprises in control group and treatment group. After implementation of EPL, the 95 % confidence interval of estimated coefficient doesn't contain zero. This indicates that compared to the control group, the biodiversity risk attention of the enterprises in treatment group has significantly increased. Although there was a downward trend in the third year, they began to increase significantly in the sixth year. This confirms the robustness of the environmental protection law in promoting biodiversity risk attention.

Furthermore, in order to test whether there is a policy expectation effect, we also conduct parallel test with external shock in previous four years. The results are shown in Figs. 6-9 indicating that there is no policy expectation effect.

3.3.2. Placebo test

In order to deal with the possibility that corporate biodiversity risk attention may be interfered with by other unobserved omitted variables, we also conduct a placebo test, randomly selecting companies among research sample as the dummy treatment group for 500 times to estimate the regression model, and the results of the placebo test are shown in Fig. 10. It can be seen that the coefficients of the false treatment group are mainly concentrated around zero, significantly deviating from the baseline regression coefficient 0.00504. This indicates misleading policy interventions do not exert considerable influence of EPL on biodiversity risk attention.

In addition, we also conduct a time placebo test, creating false policy time points to further verify the results. The results are shown in Figs. 11-14. As can be seen, the coefficient distribution is concentrated around 0, and there is a significant deviation from the basic regression coefficient.

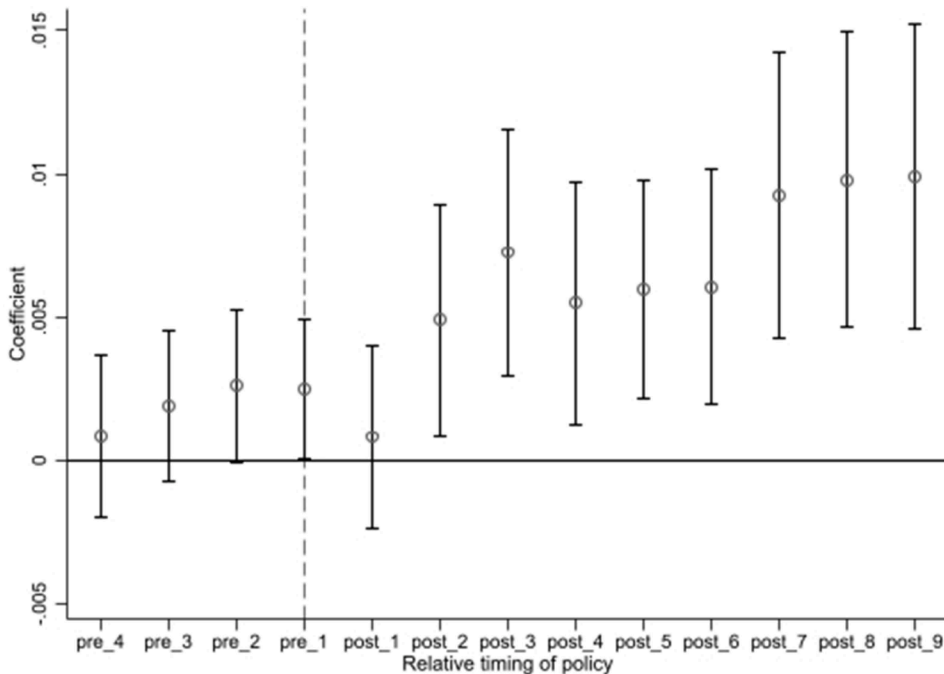


Fig. 6. Results of parallel trends test (external shock in 2014).

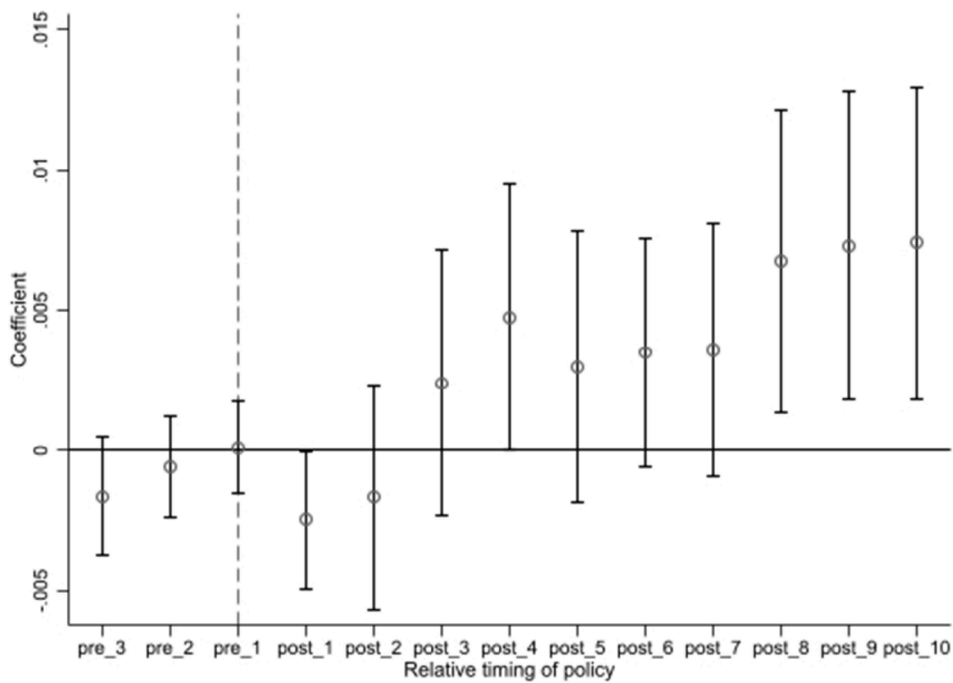


Fig. 7. Results of parallel trends test (external shock in 2013).

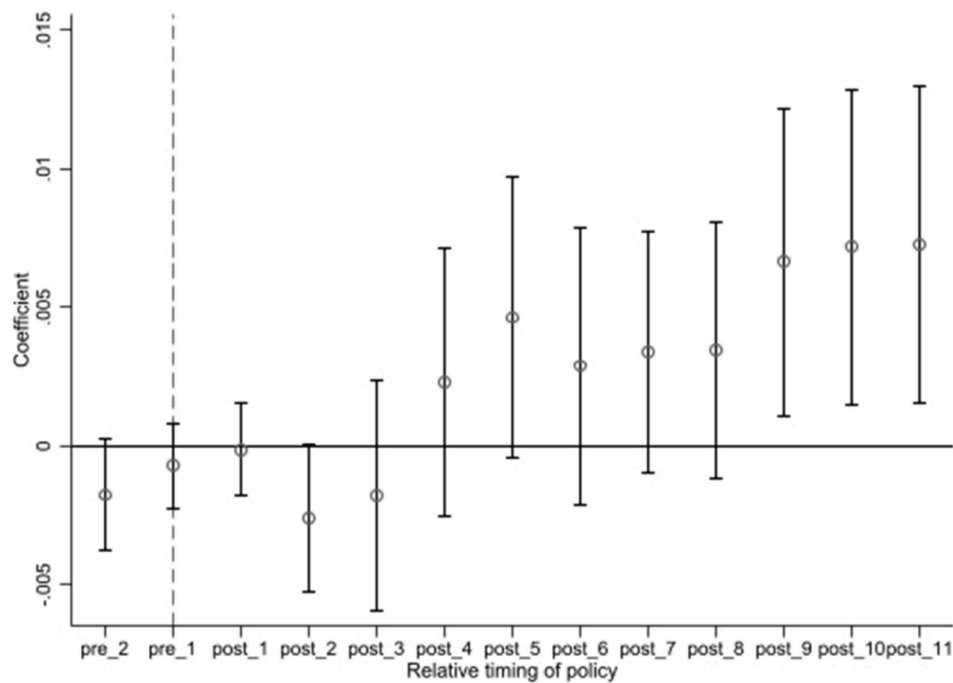


Fig. 8. Results of parallel trends test (external shock in 2012).

3.3.3. High dimensional fixed effects

To further verify the robustness of the empirical analysis, we include industry and region fixed effects in the model and re-estimate the regression model. The regression results are shown in Table 6, the estimated coefficient for interaction term DID is 0.00504 and significant at 1 % level. The implementation of EPL has still increased corporate biodiversity risk attention by 0.5 %. This further confirms the robustness of empirical results of this paper.

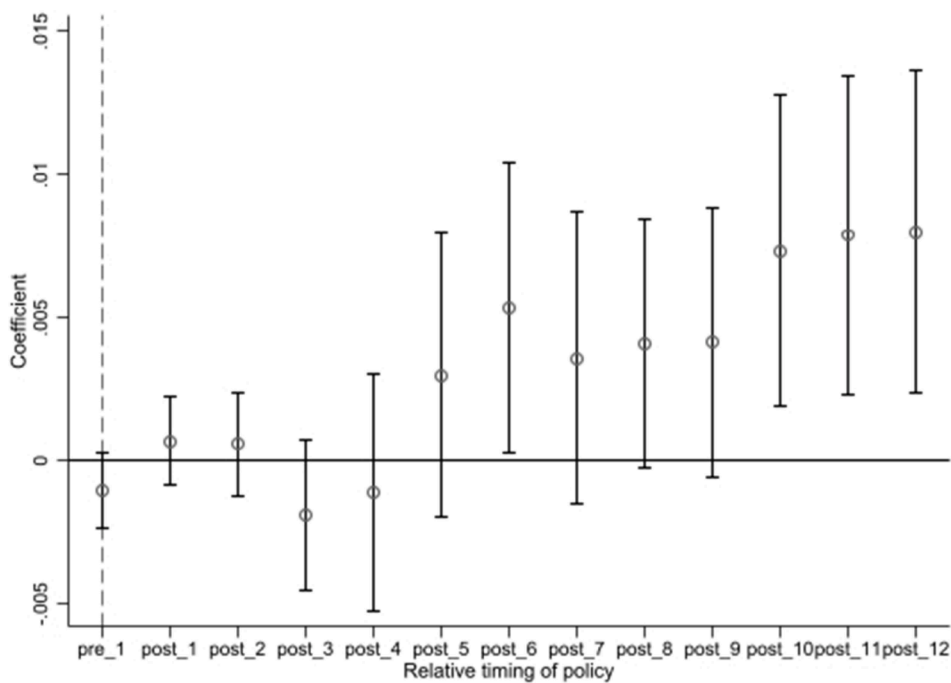


Fig. 9. Results of parallel trends test external shock in 2011).

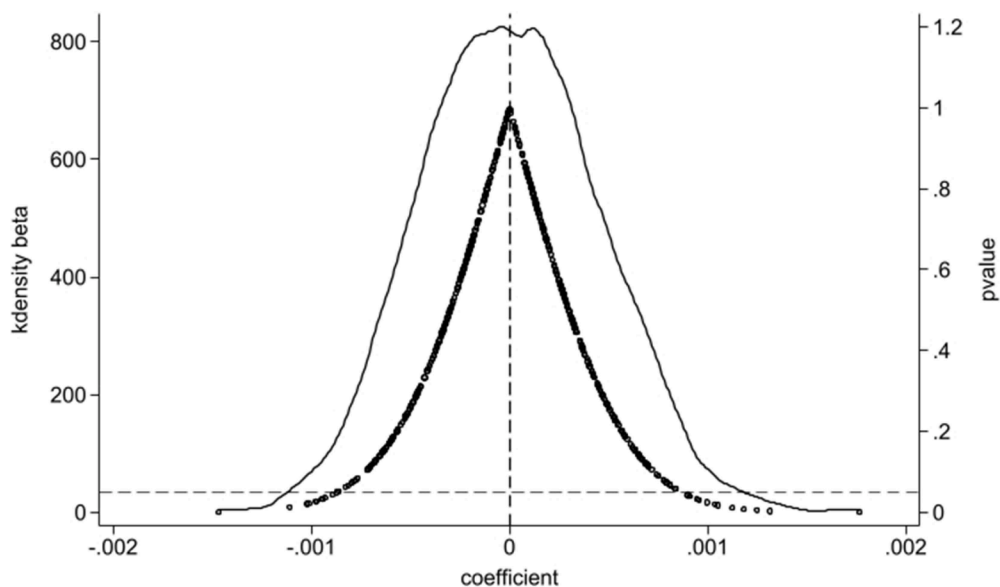


Fig. 10. Placebo test result.

3.3.4. Exclusion of other external shocks

The implementation of the Environmental Protection Tax Law in 2018 and the Biosecurity Law in 2021 may have an impact on corporate biodiversity risk attention. Therefore, we exclude the data for 2018 and 2021 to re-estimate the model and the results are shown in Table 7. This further verifies that the empirical results in this paper are robust.

In addition, we also added two interaction terms: the 2013 policy of the Yangtze River Economic Belt and the 2017 pilot policy of the national park system to the regression model. The results are shown in the Table 8, indicating that the increased attention of enterprises to biodiversity risk is indeed caused by the Environmental Protection Law.

Moreover, since text analysis methods may be influenced by corporate “green marketing” or disclosure strategies, this paper further employs carbon intensity as an alternative dependent variable to verify whether external shock environmental law can mitigate carbon

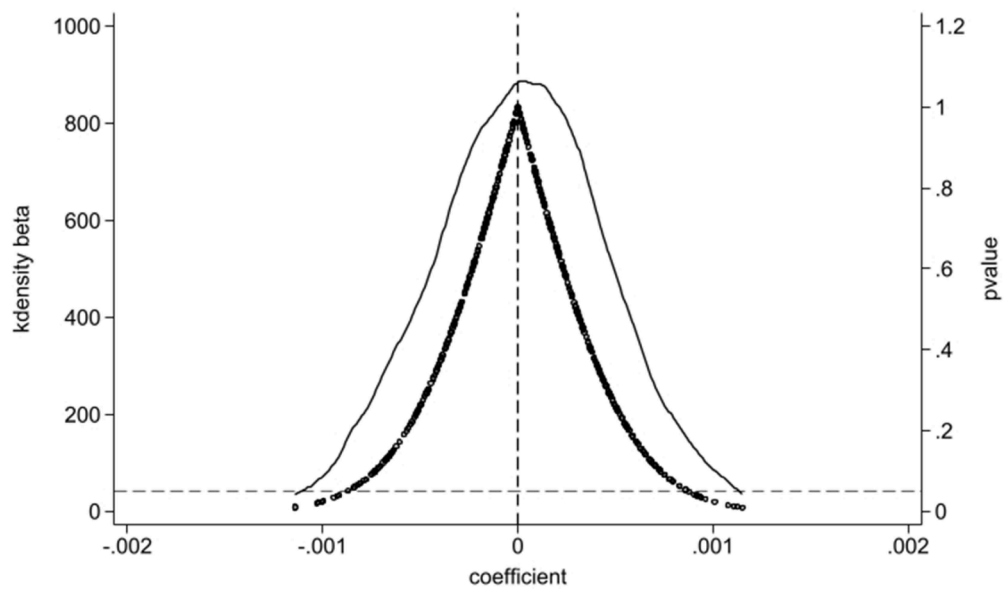


Fig. 11. Placebo test result (external shock in 2014).

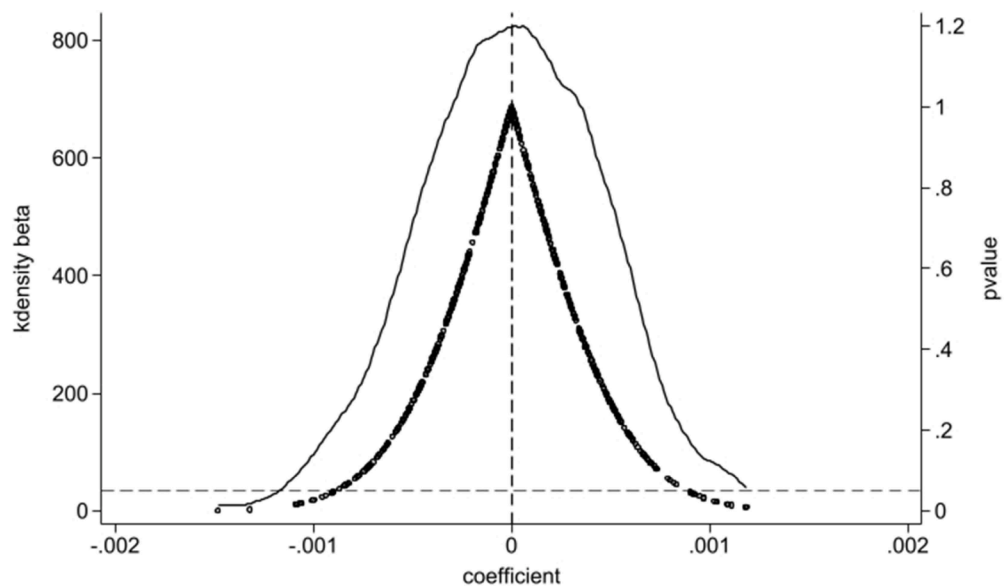


Fig. 12. Placebo test result (external shock in 2013).

intensity in biodiversity-sensitive industries. The results are shown in the [Table 9](#).

3.4. Mechanism analysis

3.4.1. Mediating effect of ESG performance

Currently, as biodiversity information disclosure is not included in ESG rating methodology, the government are also promoting the inclusion of biodiversity indicators in corporate environmental information disclosure. Enterprises will face more attention from stakeholders after obtaining higher ESG ratings. And in the future, ESG rating methodology may include biodiversity information disclosure, resulting in companies focusing on the impact of their operation activities on biodiversity risk. Therefore, companies with strong environmental protection awareness will be prompted to pay attention to biodiversity risk in addition to environmental information related to ESG ratings.

Both social responsibility practices in corporate governance and the effectiveness of corporate internal management can positively

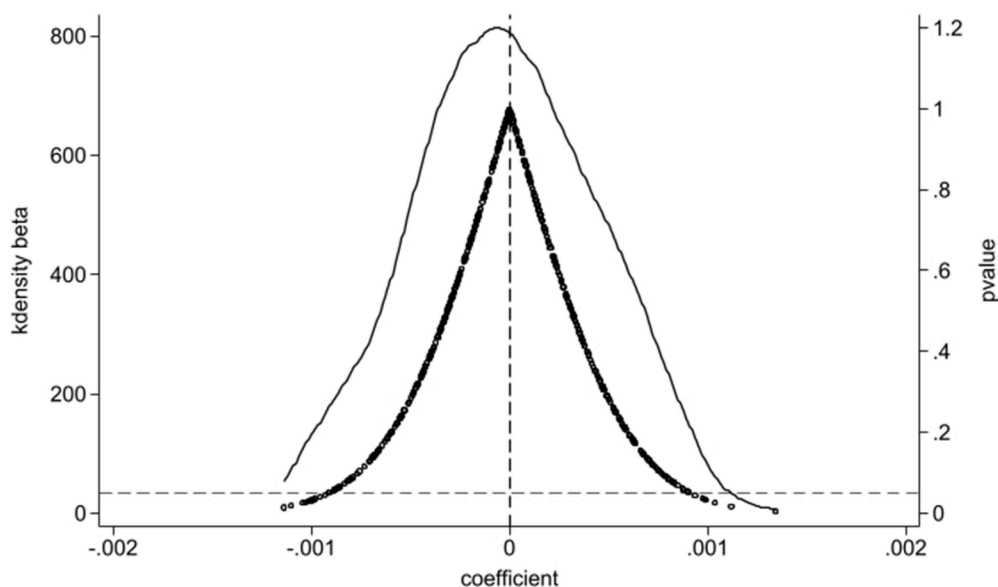


Fig. 13. Placebo test result (external shock in 2012).

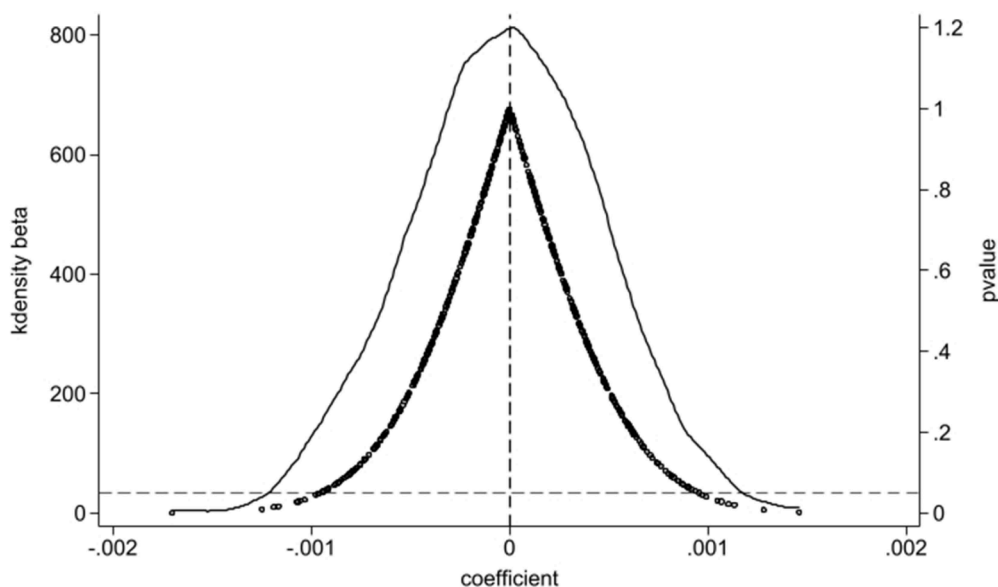


Fig. 14. Placebo test result (external shock in 2011).

influence biodiversity disclosure (Orazalin et al., 2025). (Zuo and Zhang, 2024) used textual information disclosed in annual reports to measure the green transformation of firms and found that ESG ratings can significantly contribute to the green information disclosure thereby promoting green transformation of firms.

3.4.2. Mediating effect of executives' environmental awareness

With the increasing importance of sustainable development, the environmental awareness of the senior management of enterprises has been gradually strengthened. Studies have shown that a stronger ability to perceive climate risks can drive the ESG performance of enterprises (Yin et al., 2025). The green awareness of executives can directly influence the decision-making of enterprises. Executives with high green awareness will realize the importance of environmental protection, and thus pay more attention to biodiversity risk.

Executives with green experience take a forward-looking approach in strategic decision-making, especially with regard to environmental and social responsibility risks (Chen et al., 2024). As a result, they focus more on biodiversity risk than executives without green experience. (Tu et al., 2024) argued that executives' high green awareness can improve firms' ESG performance by enhancing

the environmental information disclosure.

The results of mechanism test are shown in Table 10, the EPL does significantly contribute to corporate ESG performance with an effect magnitude of 0.017 and significant at the 1 % level, also significantly enhance the executives' environmental awareness with an effect magnitude of 0.00181 and significant at the 1 % level, which in turn promotes corporate biodiversity risk attention.

3.4.3. Bootstrap method and reverse causal relationship test

This paper further employs the bootstrap method to examine the mediating effect of ESG performance and executives' green awareness, setting the bootstrap runs to 5000 with a 95 % confidence interval. As shown in Table 11, the bias-corrected and 95 % confidence intervals for the indirect effect, direct effect, and total effect of ESG and executives' green awareness all exclude zero. This indicates the presence of a mediating effect.

To examine whether reverse causality exists, this paper employs a one-period lagged ESG performance and executives' environmental awareness as mediating variables for analysis. As shown in Table 12, the regression coefficient of DID remains statistically significant, further confirming the robustness of the mediating effect.

3.5. Heterogeneity test

3.5.1. Property right heterogeneity

Due to differences in the ability and awareness of state-owned enterprises and non-state-owned enterprises to fulfill their social responsibilities, state-owned enterprises have a stronger adjustment capabilities than non-state-owned enterprises in the face of policies issued by the government. In his 2016 instructions on the reform of state-owned enterprises, President Xi Jinping emphasized that state-owned enterprises should take the lead in innovation and green development. Therefore, state-owned enterprises will pay more attention to their own green development and pay more attention to the importance of biodiversity risks.

So, we divided the sample into SOEs and non-SOEs, the results are shown in Table 14. It shows that the biodiversity risk attention of SOEs has been significantly improved after the implementation of the EPL.

3.5.2. Carbon emissions trading pilot heterogeneity

Before the implementation of the EPL, China has set up carbon trading pilot cities in 2013, including Beijing, Shanghai, Guangdong, Fujian, Hubei, Tianjin and Chongqing. As shown in Table 13, this paper further discovers that more enterprises in the carbon emission trading pilot cities have been included in the "general ESG" funds, and the enterprises in the pilot cities attracted more green investors. This indicates that the carbon trading policy can enhance the concentration of green investors. In this atmosphere, enterprises will pay more attention to green development, thereby increasing their focus on biodiversity risk.

In this paper, the sample firms are categorized and the model is re-estimated according to whether the enterprises are located in carbon emissions trading pilot cities, and the regression results are shown in Table 14. It shows that the biodiversity risk attention of enterprises located in carbon emissions trading pilot provinces have been significantly improved after the implementation of the EPL.

Discussion

The EU's "Regulation on the Registration, Evaluation, Authorization and Restriction of Chemicals" requires enterprises to take the initiative in assuming the safety responsibilities for chemical products. It requires enterprises to submit registration to the EU Chemicals Agency; otherwise, they are not allowed to sell. This is a market-oriented "disclosure guidance" type of policy. The Chinese "Environmental Protection Law" is a "command and control" type of policy, which regulates corporate behavior through clear administrative instructions and regulatory requirements, and employs mandatory measures such as on-site inspections, information disclosure, and enforcement of violations. This governance approach has strong execution power, quick results, and a relatively centralized regulatory system. The Chinese market is large and complex, with a large number of enterprises. The policy approach based on disclosure faces problems such as poor execution by enterprises and a significant increase in regulatory difficulty. Therefore, "command and control" is more suitable for the Chinese market environment.

To better assess the ESG performance of enterprises, we suggest assigning weights to the distances between the enterprises' activities and national-level nature reserves, as well as the habitats of species with high protection levels, based on the enterprises' pollutant emissions. Then, these factors should be incorporated into the ESG evaluation indicators. Moreover, it can be observed that environmental laws have not significantly elevated the level of attention paid by enterprises to biodiversity risks. Regulatory authorities should intensify oversight and penalties to stimulate companies to voluntarily enhance their awareness of biodiversity risks.

Limitations

First, there are no established criteria for defining biodiversity-sensitive industries. This paper uses indicators such as "distance to protected areas or habitats" and "pollutant emissions" for classification.

Second, measuring corporate biodiversity risk attention through text analysis has inherent limitations due to its subjectivity.

Conclusion

This paper finds that environmental protection law can enhance corporate biodiversity risk attention, with ESG performance and

executive environmental awareness serving as mediating factors. This impact is more pronounced for state-owned enterprises and those located in carbon emissions trading pilot cities. Therefore, this paper argues that effective environmental institutions can indeed enhance corporate biodiversity risk attention by improving ESG performance and executive environmental awareness. However, it is still necessary to incorporate biodiversity risk attention into standard corporate evaluation systems at an early stage to enable environmental systems to more comprehensively influence corporate green behavior.

In this paper, only two mechanism pathways are proposed in the process of researching the influence of environmental protection law on corporate biodiversity risk attention. More mechanism paths will be explored in future research for the influencing factors of corporate biodiversity risk attention.

CRedit authorship contribution statement

Xijin Zhou: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Deqin Lin:** Writing – review & editing, Validation, Supervision, Software, Methodology, Formal analysis, Conceptualization.

Appendix

Table 1

Descriptive statistics of corporate biodiversity attention.

Biodiversity attention	Obs	Mean	Std.Dev.	P5	P50	P95
2010	1855	0.0123	0.0728	0	0	0.0378
2011	1855	0.0125	0.0750	0	0	0.0365
2012	1855	0.0144	0.0783	0	0	0.0467
2013	1855	0.0156	0.0787	0	0	0.0519
2014	1855	0.0148	0.0633	0	0.0032	0.0572
2015	1855	0.0184	0.0658	0	0.0041	0.0686
2016	1855	0.0229	0.0790	0	0.0069	0.0707
2017	1855	0.0262	0.0883	0	0.0080	0.0860
2018	1855	0.0276	0.0835	0	0.0098	0.0928
2019	1855	0.0298	0.0834	0	0.0118	0.0930
2020	1855	0.0326	0.0818	0	0.0144	0.1109
2021	1855	0.0364	0.0855	0	0.0176	0.1098
2022	1855	0.0386	0.0879	0	0.0190	0.1200
2023	1855	0.0395	0.0864	0	0.0202	0.1204

Note: P5 = 5th percentile, P50 = 50th percentile (median), P95 = 95th percentile.

Table 2

Corporate biodiversity pressure assessment standard.

Pressure factors	Direct pressure (the impact of changing land use patterns)	
	(1) Pressure on protected areas	The location of the enterprise activities is <5 km away from the national-level nature reserve.
Pressure event	(2) Pressure on the habitat of species	Distance to the habitats of species under high protection level: The location of the enterprise activities is <7 km away from the bird observation points; <3 km away from mammals, amphibians and reptiles. (Species of high protection level refer to national first-class and second-class protected wild animals, as well as critically endangered species CR and endangered species EN listed in the IUCN Red List and the China Biodiversity Red List.)
	Indirect pressure (such as the pressure exerted by pollutant emissions on biodiversity)	
Pressure event	Negative news that has been identified by relevant authorities as having a negative impact on the environment, such as environmental penalties and environmental lawsuits	

Table 3

Corporate biodiversity pressure assessment level.

Unidentified enterprise pressure	No natural reserves or species habitats were found to be close to the enterprise's activity sites, and there were no serious environmental penalties or litigation incidents.
Low-level enterprise pressure	It was discovered that there were no more than five subsidiaries of the listed company whose operation locations were close to the habitats of species with high protection levels, and no locations of the company's operations were found to be close to nature reserves, and there were no serious penalties or litigation incidents.
Medium-level enterprise pressure	It was discovered that there were more than five subsidiaries of the listed company whose operational locations were close to the habitats of species with high protection levels, and there were serious penalties or litigation incidents. However, no cases of enterprise activities being close to nature reserves were found.
High-level enterprise pressure	It was discovered that there are more than five subsidiaries of the listed company whose activity locations are close to the habitats of species with high protection levels. There are also serious penalty or lawsuit incidents, and the activity locations of these enterprises are close to national-level nature reserves.

Table 4

Variable definition and descriptive statistics.

Variables	Obs	Mean	Std.Dev	P5	P50	P95	Definition
Biodiversity risk attention (BA)	25,970	0.0244	0.0802	0	0.007	0.0834	The proportion of biodiversity key words to the total number of words in the annual report
Treat*Post	25,970	0.4879	0.4999	0	0	1	Environmental Protection Law of 2015
Total assets (TA)	25,970	22.5657	1.6098	20.4229	22.3627	25.3182	Total assets of the enterprise at the end of the year
Net profit margin on total assets (NP)	25,970	0.0306	0.7779	-0.0763	0.0297	0.1229	Net profit/total assets
Asset-liability ratio (AL)	25,970	0.4945	1.2213	0.119	0.4722	0.8405	Total liabilities/total assets
Cash and cash equivalents turnover (CT)	25,970	8.4735	32.2867	0.5881	4.2603	25.3489	Operating income/ cash at the end of the year
Revenue growth rate (RG)	25,970	2.0838	76.4787	-0.3619	0.1129	1.8667	(Operating income _t - Operating income _{t-1})/Operating income _{t-1}
Green patent application (GP)	25,970	0.8988	1.2599	0	0	3.4965	Total number of green patent applications by enterprises at the end of the year
Regional GDP	25,970	10.4439	0.7863	9.1539	10.503	11.6151	Provincial GDP
Percentage of secondary industry (SP)	25,970	0.4133	0.096	0.1926	0.4287	0.5313	Provincial secondary industry added value/ Provincial GDP
Regional Population	25,970	8.5145	0.6679	7.6102	8.6547	9.3366	Provincial year-end population

Note: P5 = 5th percentile, P50 = 50th percentile (median), P95 = 95th percentile.

Table 5

Baseline regression results.

VARIABLES	Biodiversity risk attention	
DID	0.00434** (0.00197)	0.00504** (0.00203)
TA		0.00474*** (0.000885)
NP		0.000411*** (7.10e-05)
AL		0.000299*** (0.000106)
CT		4.76e-06 (5.01e-06)
RG		1.53e-06 (1.25e-06)
GP		0.00264*** (0.000677)
GDP		0.0122** (0.00570)
SP		0.0347* (0.0198)
Population		-0.0126 (0.0108)
Constant	0.0124*** (0.000878)	-0.122 (0.101)
Individual fixed effect	Yes	Yes
Year fixed effect	Yes	Yes
Observations	25,970	25,970
Number of ID	1855	1855
R-squared	0.077	0.088

Note: 1. Numbers in parentheses are the standard errors; 2. ***, ** and * represent significant at the 1 %, 5 % and 10 % levels respectively.

Table 6

High dimensional fixed effects regression results.

VARIABLES	Biodiversity risk attention	
DID	0.00434*** (0.00105)	0.00504*** (0.00105)
Constant	0.0124*** (0.000804)	-0.122** (0.0488)
Control Variables	No	Yes
Individual fixed effect	Yes	Yes
Year fixed effect	Yes	Yes
Industry fixed effects	Yes	Yes
Region fixed effect	Yes	Yes
Observations	25,970	25,970
R-squared	0.827	0.829

Note: 1. Numbers in parentheses are the standard errors; 2. ***, ** and * represent significant at the 1 %, 5 % and 10 % levels respectively.

Table 7

Results for exclusion the impact of other external shocks.

VARIABLES	exclude 2018		exclude 2021		exclude 2018&2021	
DID	0.00453** (0.00196)	0.00512** (0.00201)	0.00397** (0.00192)	0.00470** (0.00199)	0.00414** (0.00190)	0.00476** (0.00195)
Constant	0.0124*** (0.000840)	−0.128 (0.103)	0.0124*** (0.000844)	−0.139 (0.104)	0.0124*** (0.000799)	−0.147 (0.105)
Control Variables	No	Yes	No	Yes	No	Yes
Individual fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24,115	24,115	24,115	24,115	22,260	22,260
Number of ID	1855	1855	1855	1855	1855	1855
R-squared	0.080	0.092	0.073	0.084	0.077	0.088

Note: 1. Numbers in parentheses are the standard errors; 2. ***, ** and * represent significant at the 1 %, 5 % and 10 % levels respectively.

Table 8

Results for including two external policy shocks in regression model.

VARIABLES	Biodiversity risk attention	
DID	0.00442** (0.00199)	0.00511** (0.00205)
Treat _{YZ} *Post _{YZ}	−0.000957 (0.00208)	−0.00223 (0.00202)
Treat _{NP} *Post _{NP}	−0.00250 (0.00214)	−0.00304 (0.00209)
Constant	0.0124*** (0.000804)	−0.122** (0.0488)
Controlled Variables	No	Yes
Individual fixed effect	Yes	Yes
Time fixed effect	Yes	Yes
Industry fixed effects	Yes	Yes
Region fixed effect	Yes	Yes
Observations	25,970	25,970
R-squared	0.077	0.088

Note: 1. Numbers in parentheses are the standard errors; 2. ***, ** and * represent significant at the 1 %, 5 % and 10 % levels respectively.

Note: 2. Treat_{YZ}: Set the dummy variable regions covered by the Yangtze River Economic Belt policy as 1. The remaining areas are set to 0. Post_{YZ}: Set dummy variable for the years after 2013 as 1, and the remaining years as 0. Treat_{NP}: Set the dummy variable representing the province where the national park pilot policy is implemented to 1. The remaining provinces are set to 0. Post_{NP}: Set the dummy variable for the years after 2017 to 1, and set the dummy variables for the other years to 0.

Table 9

Regression results of carbon intensity as alternative dependent variable.

VARIABLES	Carbon intensity	
DID	−0.0647** (0.0252)	−0.0611** (0.0251)
Constant	3.234*** (0.0105)	3.644*** (0.789)
Control Variables	No	Yes
Individual fixed effect	Yes	Yes
Year fixed effect	Yes	Yes
Industry fixed effects	Yes	Yes
Region fixed effect	Yes	Yes
Observations	19,018	19,018
R-squared	0.799	0.800

Table 10

Results of mechanism analysis.

VARIABLES	ESG		Executives' environmental awareness	
DID	0.0134*** (0.00355)	0.0170*** (0.00341)	0.0504*** (0.01548)	0.0539*** (0.0156)
Constant	4.217*** (0.00535)	3.881*** (0.131)	0.8429*** (0.01532)	−0.4868 (1.45687)

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Table 10 (continued)

VARIABLES	ESG		Executives' environmental awareness	
Control Variables	No	Yes	No	Yes
Individual fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Observations	25,970	25,970	25,970	25,970
Number of ID	1855	1855	1855	1855
R-squared	0.050	0.069	0.011	0.021

Note: 1. Numbers in parentheses are the standard errors; 2. ***, ** and * represent significant at the 1 %, 5 % and 10 % levels respectively.

Table 11

Results of bootstrap method.

VARIABLES	Coefficient	Bootstrap std.err.	Bias-corrected 95 %		Percentile 95 %	
			Lower	Upper	Lower	Upper
ESG(ind)	−0.0002418	0.0000997	−0.0004264	−0.0000266	−0.0004378	−0.0000395
ESG(dir)	0.0169136	0.0009934	0.0149127	0.0188457	0.0149694	0.018902
ESG(tot)	0.1667178	0.0010090	0.0146445	0.0186271	0.0147189	0.018682
Environmental awareness(ind)	0.0013146	0.0001943	0.0009806	0.0017432	0.0009556	0.0017096
Environmental awareness(dir)	0.010618	0.0009408	0.0087393	0.0124537	0.0087372	0.0124537
Environmental awareness(tot)	0.0119326	0.0009426	0.010131	0.0137958	0.010128	0.0137868

Table 12

Reverse causality test result.

VARIABLES	L1.ESG		L1.Executives' environmental awareness	
DID	0.0185*** (0.00551)	0.0102* (0.00560)	0.00692** (0.00300)	0.00598** (0.00296)
Constant	−0.0665*** (0.0234)	−0.159 (0.101)	0.0108*** (0.00145)	−0.116 (0.0970)
Control Variables	No	Yes	No	Yes
Individual fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Observations	23,996	23,996	24,115	24,115
Number of ID	1855	1855	1855	1855
R-squared	0.074	0.084	0.077	0.086

Table 13

The statistics of green investment in province.

Year	Holding shares in "general ESG" funds				Green investors			
	Trading pilot cities		Non-trading pilot cities		Trading pilot cities		Non-trading pilot cities	
	Provincial total value	Provincial average value	Provincial total value	Provincial average value	Provincial total value	Provincial average value	Provincial total value	Provincial average value
2010	166	23.71	188	7.83	274	39.14	375	15.63
2011	179	25.57	167	6.96	258	36.86	391	16.29
2012	170	24.29	158	6.58	257	36.71	342	14.25
2013	213	30.43	205	8.54	291	41.57	332	13.83
2014	272	38.86	263	10.96	357	51.00	432	18.00
2015	322	46	369	15.38	417	59.57	531	22.13
2016	329	47	387	16.13	435	62.14	595	24.79
2017	262	37.43	313	13.04	394	56.29	522	21.75
2018	273	39	260	10.83	316	45.14	382	15.92
2019	283	40.43	288	12	285	40.71	339	14.13
2020	273	39	321	13.38	253	36.14	343	14.29
2021	278	39.71	360	15	260	37.14	385	16.04
2022	267	38.14	337	14.04	223	31.86	350	14.58

Table 14

Results of heterogeneity analysis.

VARIABLES	State-owned		Non-state-owned		Trading pilot cities		Non-trading pilot cities	
DID	0.00424** (0.00206)	0.00429** (0.00196)	0.00444 (0.00333)	0.00592* (0.00354)	0.00621* (0.00356)	0.00753** (0.00382)	0.00439 (0.00291)	0.00393 (0.00282)
Constant	0.0122*** (0.00102)	−0.234* (0.138)	0.0125*** (0.00141)	−0.0580 (0.134)	0.0135*** (0.00175)	−0.1683 (0.2038)	0.0115*** (0.00085)	−0.037 (0.1086)
Controlled Variables	No	Yes	No	Yes	No	Yes	No	Yes
Individual fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(continued on next page)

Table 14 (continued)

VARIABLES	State-owned		Non-state-owned		Trading pilot cities		Non-trading pilot cities	
Observations	12,670	12,670	13,300	13,300	10,710	10,710	15,260	15,260
Number of ID	905	905	950	950	765	765	1090	1090
R-squared	0.119	0.135	0.055	0.064	0.068	0.081	0.090	0.101

Note: 1. Numbers in parentheses are the standard errors; 2. ***, ** and * represent significant at the 1 %, 5 % and 10 % levels respectively.

Data availability

Data will be made available on request.

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